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2.2 R1 (1/1 points)

While doing a homework assignment, you fit a Linear Model to your data set. You are thinking about changing the Linear Model to a Quadratic one. Which of the following is most likely true:

- ☐ Using the Quadratic Model will decrease your Irreducible Error.
- ☒ Using the Quadratic Model will decrease the Bias of your model. ✓
- ☐ Using the Quadratic Model will decrease the Variance of your model
- ☐ Using the Quadratic Model will decrease your Reducible Error

EXPLANATION

Introducing the quadratic term will make your model more complicated. More complicated models typically have lower bias at the cost of higher variance. This has an unclear effect on Reducible Error (could go up or down) and no effect on Irreducible Error.

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2.2 R2 (1/1 points)

A hypercube with side length 1 in d dimensions is defined to be the set of points $(x_1, x_2, \dots, x_d)$ such that $0 \leq x_j \leq 1$ for all $j = 1, 2, \dots, d$. Define the boundary region of the hypercube to be the set of all points such that there exists a j for which $0 \leq x_j \leq .05$ or $.95 \leq x_j \leq 1$ (i.e., it is the set of all points that have at least one dimension in the most extreme 10% of possible values). What proportion of the volume of a hypercube of dimension 50 is in the boundary region?

Please give your answer as a value between 0 and 1 with 3 significant digits. If you think the answer is 50.52%, you should say 0.505:

0.995

Answer: 0.995

EXPLANATION

We know that the volume of the whole hypercube is $1^{50} = 1$. The volume of the interior of the hypercube is $0.9^{50} = 0.005$. Thus, the volume of the boundary is $1 - 0.005 = 0.995$.

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